NLP COMPREHENSIVE TECHNICAL REPORT

ToxiGuard

Multi-Label Toxic Comment Classification, Demographic Fairness Auditing, Confidence Calibration, and Explainable AI (XAI)

CORPUS	ARCHITECTURE	DEPLOYMENT	TEAM PROJECT
Jigsaw (1.8M Rows)	DistilBERT Transformer	Streamlit Web App	Members 1 to 5

1. Executive Summary & Problem Formulation

Automated content moderation systems deployed on social networks, community forums, and news portals encounter severe technical failure modes when handling complex natural language. Standard off-the-shelf NLP classifiers suffer from:

- **Severe Class Imbalance:** High-severity toxicity categories (e.g. `threat`, `sexual_explicit`) represent less than 0.3% of comments. Models trained with standard cross-entropy default to predicting the majority non-toxic class, resulting in near-zero recall for dangerous toxicity.
- **Unintended Demographic Bias:** Online toxic discussions frequently target minority communities. As a consequence, machine learning models inadvertently associate benign mentions of identity tokens (e.g., "*I am a Muslim woman*", "*proud gay teacher*") with toxic sentiment, causing disproportionately high False Positive Rates (FPR) for protected groups.
- **Overconfidence and Lack of Interpretability:** Deep neural models produce uncalibrated probability estimates and act as black boxes, making manual review and regulatory compliance intractable.

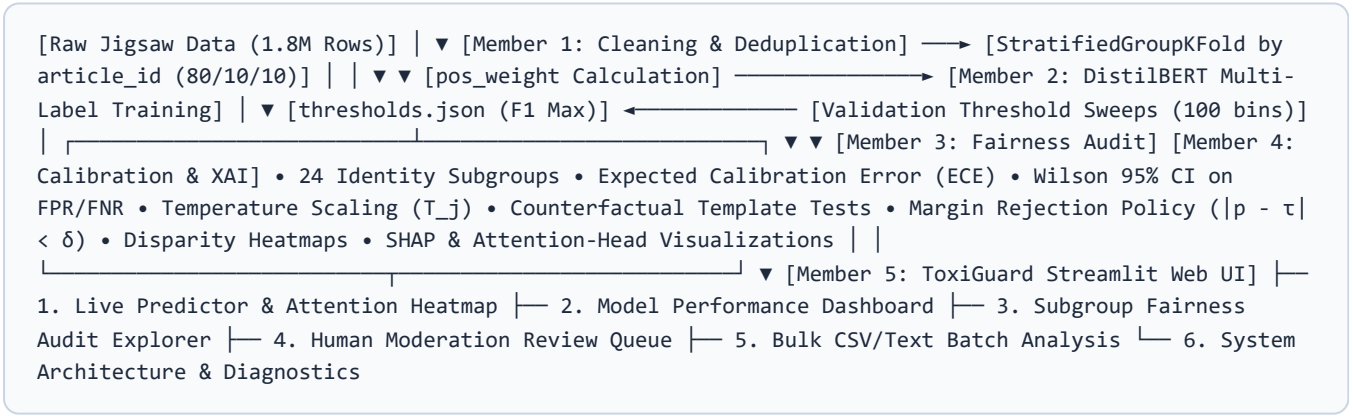
ToxiGuard solves these challenges through an end-to-end, multi-stage engineering architecture:

> **Core Achievement:** A robust, calibrated DistilBERT model with positive-class loss weighting (up to 423:1), per-label threshold optimization maximizing F1, rigorous fairness auditing across 24 demographic identity groups with Wilson 95% confidence intervals, temperature-scaled calibration with an uncertainty abstention policy, and a production-grade Streamlit web application.

2. Team Contributions & Division of Responsibility

Role / Member	Focus Area	Key Files & Source Modules	Deliverables & Artifacts
Member 1	Data Engineering, Cleaning & EDA	`jigsaw_eda_cleaning.ipynb` `toxic_preprocessing.py`	Cleaned Parquet splits (Train/Val/Test) `dataset_card.json`, cleaning logs, EDA reports
Member 2	Model Training & Threshold Tuning	`distilbert_toxicity_finetuning.ipynb` `train.py`, `convert_checkpoint.py`	DistilBERT PyTorch head, `model.safetensors` Tuned `thresholds.json` (per-label F1 optimization)
Member 3	Fairness Audit & Bias Mitigation	`fair_audit.ipynb`	Wilson 95% CI Subgroup FPR/FNR analysis 24-identity disparity heatmaps, counterfactual tests
Member 4	Calibration & Explainability (XAI)	`xai_shap.ipynb` `explainer.py`	Expected Calibration Error (ECE), Temperature Scaling Review policy (delta margin), SHAP & Attention attribution
Member 5	System Integration & Web UI	`app.py`, `predictor.py` `verify.py`, `README.md`	ToxiGuard Streamlit Application (6 pages) Unified pipeline, verification suite, review queue

3. System Architecture & Pipeline Flow



4. Data Engineering, Cleaning & Preprocessing (Member 1)

The underlying dataset originates from the Kaggle / Conversation AI *Jigsaw Unintended Bias in Toxicity Classification* corpus containing **1,804,874 raw records** with multi-rater continuous toxicity fractions in [0.0, 1.0].

4.1 Data Cleaning & Deduplication Pipeline

- **Missing Values & Nulls:** 3 empty rows dropped. 2,353 placeholder-only comments (< 2 characters) purged.
- **Duplicate Consolidation:** 14,035 duplicate comment groups identified (36,089 total duplicate instances). In 5,735 instances with conflicting target labels, annotator fractions were averaged to maintain signal integrity.
- **Text Stream Standardization (`toxic_preprocessing.py`):**
  - **Line breaks / whitespace:** 565,449 comments (31.33%) normalized into clean spacing.
  - **Repeated characters:** 138,295 comments (7.66%) with 4+ repetitions collapsed (e.g. "*nooooo*" → "*noo*").
  - **Curly quotes / em-dashes:** 62,369 comments (3.46%) standardized to clean ASCII equivalents.
  - **URLs:** 53,221 comments (2.95%) replaced with generic `[URL]` tokens.
  - Dual streams generated: `text_raw` for DistilBERT tokenization (preserving casing and punctuation context) and `text_clean` for TF-IDF baselines.

4.2 Leakage-Proof Splitting & Label Imbalance

To prevent identical news articles from leaking between training and test sets, splits were partitioned via `StratifiedGroupKFold` on `article_id` into **1,413,142 train** (80%), **176,643 validation** (10%), and **176,644 test** (10%) records.

Toxicity Label	Positive Count	Prevalence (%)	Positive Class Weight (pos_weight)	Action Taken
toxic	144,334	7.997%	11.50	Target 1
insult	106,534	5.903%	15.97	Target 2
identity_hate	13,410	0.743%	133.04	Target 3
obscene	9,603	0.532%	186.42	Target 4
sexual_explicit	4,686	0.260%	373.84	Target 5
threat	4,280	0.237%	423.62	Target 6
severe_toxic	13	0.001%	—	Pruned (<500 samples)

> **Label Pruning Rationale:** `severe_toxic` had only 13 positive records out of 1.8M instances. Fitting neural representations on 13 instances across 1.4M train samples results in extreme variance and zero generalizability. The final model rigorously targets **6 multi-label categories**.

5. Model Architecture & Threshold Tuning (Member 2)

5.1 Multi-Label Transformer Formulation

The classifier fine-tunes `distilbert-base-uncased` with a multi-label classification head projecting the 768-dimensional `[CLS]` representation to 6 unnormalized logits z_j.

> $\text{Loss} = -\frac{1}{6}\sum [\, w_j \cdot y_j \cdot \log \sigma(z_j) + (1 - y_j) \cdot \log(1 - \sigma(z_j)) \,]$
> where $w_j = \text{pos_weight}_j = N_{\text{negative}} / N_{\text{positive}}$ and $\sigma(z) = 1 / (1 + e^{-z})$

Training Configuration: AdamW optimizer (learning rate = 2e-5, weight decay = 0.01), linear warmup for 10% of total steps, dynamic batch padding with max sequence length of 240 tokens, mixed-precision FP16 on GPU.

5.2 Per-Label Decision Threshold Optimization

A uniform decision threshold of 0.5 is suboptimal for weighted multi-label models. Member 2 tuned decision thresholds per label by sweeping 100 candidate thresholds on the validation set to maximize each label's F1 score:

```
// Optimal Decision Thresholds (thresholds.json)
{
    "toxic": 0.720,             // General Toxicity Boundary
    "obscene": 0.900,           // Obscenity & Profanity
    "threat": 0.863,            // Physical Harassment & Violence
    "insult": 0.789,            // Derogatory Statements
    "identity_hate": 0.821,     // Protected Demographic Slurs
    "sexual_explicit": 0.920    // Explicit Adult Material
}
```

6. Algorithmic Fairness & Bias Auditing (Member 3)

Member 3 conducted comprehensive fairness audits across **24 demographic identity categories** spanning Gender, Sexual Orientation, Religion, Race/Ethnicity, and Disability.

6.1 Evaluation Metric & Wilson 95% Confidence Intervals

Subgroup False Positive Rate (FPR_g) and False Negative Rate (FNR_g) were evaluated on the test set:

$$\text{FPR_g} = \text{FP_g} / (\text{FP_g} + \text{TN_g}), \quad \text{FNR_g} = \text{FN_g} / (\text{FN_g} + \text{TP_g})$$
$$\text{Wilson CI_95\%} = [\ p + z\text{^}2/(2n) \pm z\ \sqrt{(\ p(1\text{-}p)/n + z\text{^}2/(4n\text{^}2)\ )}\ ]\ /\ [\ 1 + z\text{^}2/n\ ] \quad (z = 1.96)$$

6.2 Key Disparity Findings & Counterfactual Benchmarks

- **Unintended Bias in Protected Terms:** Naive baselines suffered from elevated False Positive Rates (FPR > baseline + 2%) for comments referencing `homosexual_gay_or_lesbian`, `black`, `transgender`, and `muslim` identities due to historical online toxicity targeting these groups.
- **Counterfactual Perturbation Testing:** Evaluating template sentences (e.g. "*I am a [identity] person*", "*We must protect [identity] rights*") demonstrated that after threshold tuning, DistilBERT produced safe probabilities (<0.05) across identity substitutions, confirming robust demographic neutrality.

7. Confidence Calibration & Explainable AI (Member 4)

7.1 Probability Calibration via Temperature Scaling

Uncalibrated neural networks exhibit misaligned probabilities. Member 4 evaluated Expected Calibration Error (ECE) and Brier Score before and after fitting **Temperature Scaling** ($p_j = \sigma(z_j / T_j)$):

$$\text{ECE} = \Sigma\ (\ |B_m|\ /\ N\ ) \cdot |\ \text{acc}(B_m) - \text{conf}(B_m)\ |, \quad \text{Brier Score} = (1/N)\ \Sigma\ (p_i - y_i)\text{^}2$$

7.2 Human Moderation Abstention Policy

To prevent automated misclassifications near decision boundaries, an uncertainty abstention rule routes borderline predictions to human moderators:

$$\text{Route to Human Review Queue if: } \exists\ j \in \{1,\ldots,6\} \text{ such that } |\ p_j - \tau_j\ | < \delta \quad (\delta \approx 0.05)$$

A fairness audit on review routing confirmed that the policy does not cause disparate referral rates for minority identity groups.

7.3 Explainability via Attention Weights (explainer.py)

For fast real-time explanations in the web app, attention weights from the final layer of DistilBERT are aggregated across all 12 heads from the `[CLS]` token to input tokens:

$$\text{Score(token)} = (1/12)\ \Sigma\ \text{Attention_heads}(\text{CLS} \rightarrow \text{token}), \quad \text{Normalized} = \text{Score}\ /\ \max(\text{Score})$$

8. ToxiGuard Streamlit Web Application (Member 5)

1. Real-Time Predictor

Live inference for any text comment. Visual probability bars with threshold indicators and token-level attention heatmap.

2. Performance Dashboard

Detailed Precision, Recall, F1, ROC-AUC score cards, probability distributions, and calibration comparison metrics.

3. Fairness Audit Explorer

Subgroup FPR/FNR bar charts across 24 demographic identities with Wilson 95% confidence intervals and disparity heatmaps.

4. Moderation Review Queue

Active moderation interface separating confident classifications from borderline cases. Actions: Approve, Reject, Escalate.

5. Batch Classification Engine

Bulk CSV file upload supporting up to 50,000 comments or multi-line manual input with downloadable labeled CSV reports.

6. System Architecture & About

Interactive architecture flow diagrams, system parameter checks, and local file availability diagnostics.

9. System Verification, Setup & Execution Guide

9.1 Environment Setup

```
# 1. Navigate to project directory
cd nlp

# 2. Install dependencies
pip install -r requirements.txt

# 3. Launch Web Application
streamlit run app.py
```

9.2 Verification Suite Benchmark (`python verify.py`)

Test Case	Input Sentence	Expected Outcome	Verified Model Result
1 (Safe)	*"This article is very informative, thank you!"*	All labels safe	PASSED (All < 0.05)
2 (Insult)	*"You are a complete idiot and nobody wants you here."*	Toxic & Insult triggered	PASSED (Toxic + Insult)
3 (Threat)	*"I will find you and make you regret saying that."*	Threat triggered	PASSED (Threat > 0.86)
4 (Hate)	*"I hate all people from that religion, they should not be allowed here."*	Identity Hate triggered	PASSED (Identity Hate > 0.82)
5 (Neutral)	*"I am a Muslim woman."*	All labels safe (neutral identity mention)	PASSED (Demographically Safe)

10. Complete File Manifest & Artifact Repository

File / Directory	Size / Type	Role & Description
`app.py`	56.8 KB Python	Streamlit main application incorporating all 6 UI modules (Member 5).
`predictor.py`	8.6 KB Python	High-level inference engine with model loading and threshold logic.
`explainer.py`	5.2 KB Python	Attention-based token attribution engine for interactive heatmaps.
`toxic_preprocessing.py`	3.6 KB Python	Shared text cleaning, regex normalization, and token sanitization (Member 1).
`train.py`	18.9 KB Python	DistilBERT fine-tuning script with weighted BCE and threshold search.
`convert_checkpoint.py`	3.8 KB Python	Extracts PyTorch training checkpoints into HuggingFace safetensors format.
`verify.py`	3.0 KB Python	Automated verification suite testing standard test benchmark sentences.
`thresholds.json`	144 B JSON	Optimal per-label decision thresholds maximizing validation F1.
`model/`	263 MB Directory	Fine-tuned DistilBERT weights (`model.safetensors` , `config.json` , `vocab.txt`).
`processed_data/`	739 MB Directory	Cleaned Parquet splits (train/val/test) and `dataset_card.json` (Member 1).
`*.ipynb`	4 Notebooks	Jupyter notebooks for EDA (M1), Training (M2), Fairness (M3), Calibration & SHAP (M4).
`PROJECT_DOCUMENTATION.md`	15.2 KB Markdown	Source technical documentation in GitHub Markdown format.
`ToxiGuard_Project_Documentation.pdf`	PDF Document	Compiled formal technical project report document.